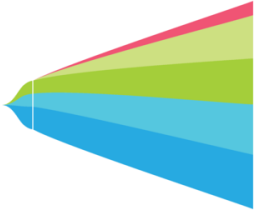


REPORTING	PHYSICIAN	
Report Date: JAN-18-2025	Susanne Arnold	
Receipt Date: JAN-11-2025	Account: University of KY HNR	
Collection Date: JAN-09-2025	Address: 800 Rose St, Ben Roach Bldg CC 222, Lexington, KY, 40508, United States	
Specimen: Blood	Ph: (859) 257-4488 Fax: (859) 257-0332	
Status: FINAL	Additional Recipient: N/A	Complete Tumor Response Map on page 3

This content is provided as a professional service and has not been reviewed or approved by the FDA

Summary of Detected Somatic Alterations, Immunotherapy Biomarkers & Associated Treatment Options

KEY ✔ Approved in indication ⚡ Approved in other indication ✖ Lack of response

Detected Alteration(s) / Biomarker(s)	Associated FDA-approved therapies	Clinical trial availability (see page 4)	% cfDNA or Amplification
ALK L1196Q	✖ Alectinib, Crizotinib	Yes	3.2%
ALK C1156Y	✖ Ceritinib, Crizotinib	Yes	2.4%
ALK G1202L	✖ Brigatinib, Crizotinib	No	0.6%
EML4-ALK Fusion	✔ Ensartinib, Lorlatinib	Yes	2.6%
CDKN2A R80*	None	Yes	2.9%

Variants of Uncertain Clinical Significance
ALK-EML4 Fusion (4.7%)
The functional consequences and/or clinical significance of alterations are unknown. Relevance of therapies targeting these alterations is uncertain.

Comments
Reported by: HM8

Biomarker	Additional Details
MSI-High	NOT DETECTED

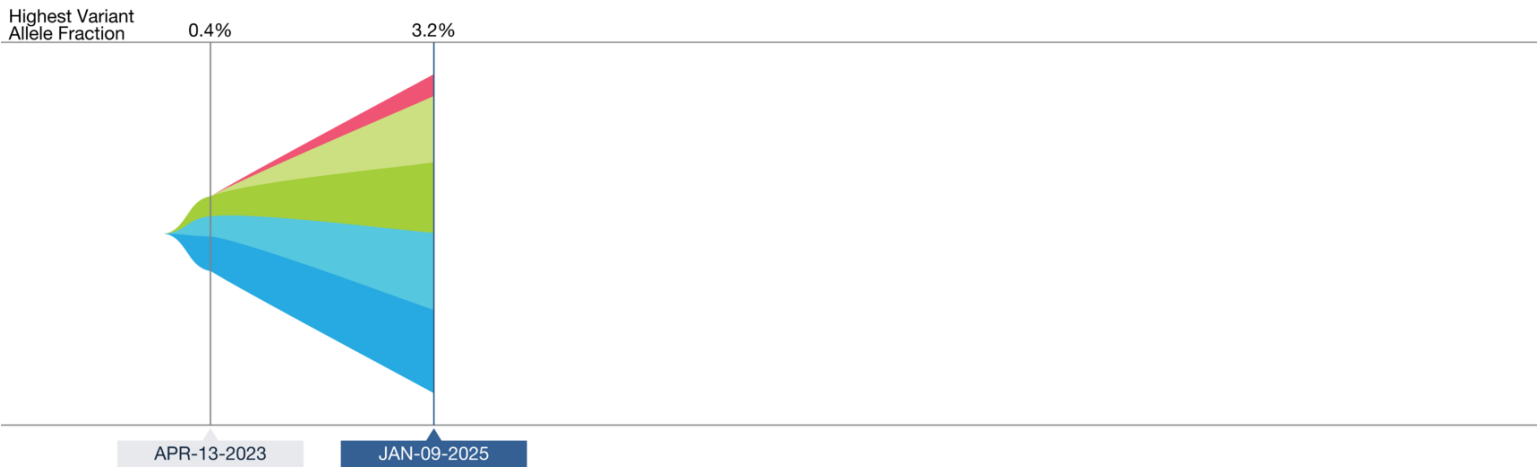
We evaluated this sample for 74 genes, including the following guideline-recommended genes for NSCLC

EGFR(T790M and others)	ALK	ROS1	BRAF	MET	ERBB2(HER2)	RET	NTRK	KRAS
------------------------	-----	------	------	-----	-------------	-----	------	------

This content is provided as a professional service and has not been reviewed or approved by the FDA

Guardant360 Tumor Response Map

The Guardant360 laboratory developed test (LDT) Tumor Response Map illustrates the variant allele fraction (% cfDNA) of observed somatic variants at each sample submission. Amplifications are not plotted, and only the first and last five test dates are plotted. Please see the Physician Portal (portal.guardanthealth.com) for the Tumor Response Map with all test dates.



Detected Alteration(s) / Biomarker(s)	% cfDNA or Amp	Alteration Trend	
ALK L1196Q	3.2%		
CDKN2A R80*	2.9%		
EML4-ALK Fusion	2.6%		
ALK C1156Y	2.4%		
ALK G1202L	0.6%		
ALK-EML4 Fusion	4.7%		Variants of Uncertain Clinical Significance §

The table above annotates the variant allele fraction (% cfDNA) detected in this sample, listed in descending order.
§ See definitions section for more detail

Available Clinical Trials (within the same state as the ordering physician)

There may be additional trials not listed here. Visit: portal.guardanthealth.com or email clientservices@guardanthealth.com with A1262117 in the subject line of the email, for additional trials.

Alteration	Trial ID / Contact	Title	Phase	Site(s)
CDKN2A R80*	NCT03994796 Priscilla Brastianos, MD, pbrastianos@partners.org,617-724-1074	Genetic Testing in Guiding Treatment for Patients with Brain Metastases	Phase 2	Lexington, KY (2)
	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
ALK L1196Q	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
ALK C1156Y	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
EML4-ALK Fusion	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			

More clinical trial options available at portal.guardanthealth.com

Definitions

Variants of Uncertain Clinical Significance: The functional consequences and/or clinical significance of alterations are unknown. Relevance of therapies targeting these alterations is uncertain.

Fusion: Fusion events are gene rearrangements that fuse two previously distinct genes into a single genetic unit. Guardant360 detects fusions in the genes listed in Table 1.

***Nonsense mutation:** A point mutation that results in a premature stop codon.

Interpretation

Somatic alterations were detected in the circulating cell-free DNA isolated from this patient's blood specimen. These genomic alterations are cancer-associated somatic variants, some of which have been associated with either increased or reduced clinical response to specific treatments. The percentage of altered cell-free DNA circulating (% cfDNA) in blood is related to the unique tumor biology of each patient. Factors that may affect the % cfDNA of detected somatic alterations include tumor growth, turn over, size, heterogeneity, vascularization, disease progression, and treatment.

Method and Limitations

Guardant360 sequences 74 cancer-associated genes to identify somatic alterations. Cell-free DNA (cfDNA) is extracted from plasma, enriched for targeted regions, and sequenced using the Illumina platform and hg19 as the reference genome. All exons are sequenced in some genes; only clinically significant exons are sequenced in other genes. The types of genomic alterations detected by Guardant360 include single nucleotide variants, gene amplifications, fusions, short insertions/deletions (longest detected, 70 base pairs), and splice site disrupting events (see Table 1). Microsatellite Instability (MSI) is assessed for all cancer types by evaluating somatic changes in the length of repetitive sequences on the Guardant360 panel. A "Not Detected" result in samples where the highest % cfDNA is < 0.2% is an inconclusive result because it does not preclude MSI-High status in tissue. MSI status is currently not reported for earlier panel versions. This version of the Guardant360 test is not validated for the detection of other types of genomic alterations, such as complex rearrangements or gene deletions. Certain sample or variant characteristics, such as low cfDNA concentration, may result in reduced analytic sensitivity. Guardant360 cannot discern the source of circulating cfDNA, and for some variants in the range of ~40 to 60% cfDNA, the test cannot easily distinguish germline variants from somatic alterations. Guardant360 is not validated for the detection of germline or de novo variants that are associated with hereditary cancer risk. Tissue genotyping should be considered when plasma genotyping is negative, if clinically appropriate.

Table 1: Genes on the Guardant360 Panel

Guardant360 reports single nucleotide variants, splice site mutations, and insertion and deletion variants (indels) in all clinically relevant exons in 74 genes and reports other variant types in select genes as indicated below.

AKT1	ALK #	APC	AR †	ARAF	ARID1A	ATM	BRAF †	BRCA1
BRCA2	CCND1 †	CCND2 †	CCNE1 †	CDH1	CDK12	CDK4 †	CDK6 †	CDKN2A
CTNNB1	DDR2	EGFR †	ERBB2 †	ESR1	EZH2	FBXW7	FGFR1 †	FGFR2 † #
FGFR3 #	GATA3	GNA11	GNAQ	GNAS	HNF1A	HRAS	IDH1	IDH2
JAK2	JAK3	KIT †	KRAS †	MAP2K1	MAP2K2	MAPK1	MAPK3	MET †
MLH1	MPL	MTOR	MYC †	NF1	NFE2L2	NOTCH1	NPM1	NRAS
NTRK1 #	NTRK3	PDGFRA †	PIK3CA †	PTEN	PTPN11	RAF1 †	RB1	RET #
RHEB	RHOA	RIT1	ROS1 #	SMAD4	SMO	STK11	TERT ‡	TP53
TSC1	VHL							

‡ Guardant360 reports alterations in the promoter region of this gene. # Guardant360 reports fusion events involving this gene for all known gene partners.
† Guardant360 reports amplifications of this gene.

About the Test

The Guardant360 assay was developed and its performance characteristics were determined by Guardant Health, Inc. This test has not been cleared or approved by the U.S. Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary. This test may be used for clinical purposes and should not be regarded as investigational or for research only. Guardant Health's clinical reference laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high complexity clinical laboratory testing. The laboratory report should be interpreted in the context of other clinical information and laboratory, pathology, and imaging studies by a qualified medical professional prior to initiating or changing a patient's treatment plan. The selection of any, all, or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) is entirely at the discretion of the treating medical professional. Drug and trial information are based on the diagnosis written on the submitted test request form; this information is not based on any supplemental information provided by the requesting medical professional, including pathology reports or other molecular studies. Some drugs listed in this report may not be approved or cleared by the FDA for the indicated use. Guardant Health makes no endorsement, express or implied, of any product, physician, or procedure contained in this report. This report makes no promises or guarantees that a particular medication will affect (or not affect) the clinical outcome of any patient.

Testing performed at: Guardant Health
Laboratory Director: Martina Lefterova, MD PhD | CLIA ID: 05D2070300 | CAP #: 8765297 | 505 Penobscot Drive, Redwood City, CA, 94063, USA

Additional information is available

Any therapeutic annotations are based on publicly available information. This information is described in the "Detailed Therapy Results" and "Relevance of Detected Alterations" sections.

Visit portal.guardanthealth.com or email clientservices@guardanthealth.com with A1262117 in the subject line of the email for:

- Additional clinical trials
- Detailed Therapy Results
- Relevance of Detected Alterations
- References

If you would like to receive this additional information with every Guardant360 report, please call client services at [855.698.8887](tel:855.698.8887) to opt-in.

Additional information begins on the next page.

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
CDKN2A R80*	NCT03994796 Priscilla Brastianos, MD, pbrastianos@partners.org,617-724-1074	Genetic Testing in Guiding Treatment for Patients with Brain Metastases	Phase 2	Allentown, PA; Deerfield Beach, FL; Minneapolis, MN; Rochester, NY; Midlothian, VA; Worcester, MA; Colorado Springs, CO; Berlin, VT; Neptune, NJ; Boston, MA; Richmond, VA; Summit, NJ; Oklahoma City, OK; Edina, MN; Bremerton, WA; Bethlehem, PA; Orange, CA; Jacksonville, FL; Kennewick, WA; Pennington, NJ; Kearney, NE; Jackson, MS; Rochester, MN; Miami, FL; San Diego, CA; Burlington, VT; La Jolla, CA; Lake Success, NY; Salt Lake City, UT; Coral Gables, FL; New York, NY (2); Lexington, KY (2); WI (19); IA (13); OH (22); ID (6); MI (46); OR (6); IL (19); MT (7); NC (6)
	NCT05238922 Incyte Corporation Call Center (US), medinfo@incyte.com,1.855.463.3463	Study of INCB123667 in Subjects With Advanced Solid Tumors	Phase 1	Los Angeles, CA; Duarte, CA; Madison, WI; Philadelphia, PA; Atlanta, GA; Pittsburgh, PA; Irvine, CA; Lone Tree, CO; Fort Worth, TX; Hackensack, NJ; Fairfax, VA; Miami Beach, FL; New York, NY (2); Puerto Rico; Netherlands (2); Japan (5); United Kingdom (3); Italy (5); France (4); Switzerland (3)
	NCT05252416 Blueprint Medicines, medinfo@blueprintmedicines.com,617-714-6707	(VELA) Study of BLU-222 in Advanced Solid Tumors	Phase 1 /Phase 2	Oklahoma City, OK; Philadelphia, PA; Stanford, CA; Chicago, IL; Baltimore, MD; San Francisco, CA; Little Rock, AR; Houston, TX; Detroit, MI; Chapel Hill, NC; Sarasota, FL; Salt Lake City, UT; Nashville, TN; Charlottesville, VA; New York, NY (3); Boston, MA (2); United Kingdom; Italy (3)
	NCT05999994 There may be multiple sites in this clinical trial. 1-877-CTLILLY (1-877-285-4559) or, Clinicaltrials.gov@lilly.com,1-317-615-4559	A Master Protocol (LY900023) That Includes Several Clinical Trials of Drugs for Children and Young Adults With Cancer	Phase 2	Denver, CO; Milwaukee, WI; Minneapolis, MN; Phoenix, AZ; New York, NY; Chicago, IL; Columbus, OH; Orange, CA; Portland, OR; Jacksonville, FL; Los Angeles, CA; Providence, RI; Seattle, WA; Boston, MA; Atlanta, GA; Pittsburgh, PA; Fort Myers, FL; Dallas, TX; Washington, DC; Birmingham, AL; Ann Arbor, MI; Kansas City, MO; Houston, TX (2); Belgium; Netherlands (2); Japan (3); United Kingdom (5); Italy (7); France (4); Australia (4); Germany (3); Spain (13)
	NCT06120283 Study Director,clinicaltrials@beigene.com, 1.877.828.5568	BGB-43395 Alone or as Part of Combination Therapies in Participants With Breast Cancer and Other Advanced Solid Tumors	Phase 1	Denver, CO; Houston, TX; Detroit, MI; Saint Louis, MO; Irving, TX; Lake Mary, FL; Columbus, OH; San Antonio, TX; Nashville, TN; Durham, NC; Moldova, Republic of; New Zealand (2); Japan (3); China (7); Korea, Republic of (8); Brazil (11); Malaysia (4); Thailand (3); France (8); Australia (8)
ALK L1196Q	NCT03093116 BMS Study Connect Contact Center www.BMSStudyConnect.com,Clinical.	A Study of Repotrectinib (TPX-0005) in Patients With Advanced Solid Tumors Harboring ALK, ROS1, or NTRK1-3	Phase 1 /Phase 2	Appleton, WI; Saint Louis, MO; Baltimore, MD; Memphis, TN; Peoria, IL; Bolivar, MO; Fairfax, VA; Ann

This content is provided as a professional service and has not been reviewed or approved by the FDA

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
	Trials@bms.com,855-907-3286	Rearrangements		Arbor, MI; Goldsboro, NC; Philadelphia, PA; Chicago, IL; Columbus, OH; New Brunswick, NJ; Columbus, GA; Saint Paul, MN; Seattle, WA; Kingwood, TX; Cleveland, OH; Athens, GA; Hollywood, FL; Cincinnati, OH; Canton, OH; Dallas, TX; Tampa, FL; Houston, TX (2); Detroit, MI (2); Boston, MA (3); Washington, DC (2); Aurora, CO (2); New York, NY (3); CA (7); Korea, Democratic People's Republic of; Denmark; Singapore (2); Hungary (2); Hong Kong (2); Japan (9); United Kingdom (5); Spain (8); Canada (5); Netherlands (2); Belgium (2); China (30); Taiwan (3); Poland (5); Korea, Republic of (10); Italy (6); France (8); Australia (3); Germany (4)
	NCT05257512 Yinghui Sun, PhD,yhsun@centaurusbio.com,86-10-88858616	Phase I/II Study of SY-3505 in Patients With ALK-positive Advanced Non-small Cell Lung Cancer	Phase 1 /Phase 2	China
	NCT05341583 Chang Li Wang, Ph.D, wangchangli@tjmuch.com,86-22-23340123 x3180	Ensartinib as Adjuvant Treatment in Anaplastic Lymphoma Kinase (ALK) Positive Non-small Cell Lung Cancer	Phase 3	China
	NCT05384626 Tina Kehrig,clinicaltrials@nuvalent.com, 857-357-7000	A Study of NVL-655 in Patients With Advanced NSCLC and Other Solid Tumors Harboring ALK Rearrangement or Activating ALK Mutation (ALKOVE-1)	Phase 1 /Phase 2	Denver, CO; Philadelphia, PA; Saint Louis, MO; Stanford, CA; Chicago, IL; Columbus, OH; Sacramento, CA; Orange, CA; Baltimore, MD; Durham, NC; Seattle, WA; Detroit, MI; Miami, FL; Atlanta, GA; Washington, DC; Nashville, TN; New York, NY (2); Boston, MA (2); Belgium; Singapore (2); Japan (5); United Kingdom (4); Switzerland (2); Spain (5); Canada (4); Netherlands (2); Taiwan (3); Korea, Republic of (4); Italy (7); France (4); Australia (3); Germany (3)
	NCT06007937 Gregory Riely, MD, PhD,rielyg@mskcc.org,646-608-3913	A Study of Lorlatinib in Combination With Ramucirumab in People With Lung Cancer	Phase 1 /Phase 2	Harrison, NY; Montvale, NJ; Rockville Centre, NY; Commack, NY; New York, NY; Middletown, NJ
ALK C1156Y	NCT03093116 BMS Study Connect Contact Center www.BMSStudyConnect.com,Clinical.Trials@bms.com,855-907-3286	A Study of Repotrectinib (TPX-0005) in Patients With Advanced Solid Tumors Harboring ALK, ROS1, or NTRK1-3 Rearrangements	Phase 1 /Phase 2	Appleton, WI; Saint Louis, MO; Baltimore, MD; Memphis, TN; Peoria, IL; Bolivar, MO; Fairfax, VA; Ann Arbor, MI; Goldsboro, NC; Philadelphia, PA; Chicago, IL; Columbus, OH; New Brunswick, NJ; Columbus, GA; Saint Paul, MN; Seattle, WA; Kingwood, TX; Cleveland, OH; Athens, GA; Hollywood, FL; Cincinnati, OH; Canton, OH; Dallas, TX; Tampa, FL; Houston, TX (2); Detroit, MI (2); Boston, MA (3); Washington, DC (2); Aurora, CO (2); New York, NY (3); CA (7); Korea, Democratic People's Republic of; Denmark; Singapore (2); Hungary (2); Hong Kong (2); Japan (9); United Kingdom (5); Spain (8); Canada (5); Netherlands (2); Belgium

This content is provided as a professional service and has not been reviewed or approved by the FDA

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
				(2); China (30); Taiwan (3); Poland (5); Korea, Republic of (10); Italy (6); France (8); Australia (3); Germany (4)
	NCT05257512 Yinghui Sun, PhD,yhsun@centaurusbio.com,86-10-88858616	Phase I/II Study of SY-3505 in Patients With ALK-positive Advanced Non-small Cell Lung Cancer	Phase 1 /Phase 2	China
	NCT05341583 Chang Li Wang, Ph.D, wangchangli@tjmuch.com,86-22-23340123 x3180	Ensartinib as Adjuvant Treatment in Anaplastic Lymphoma Kinase (ALK) Positive Non-small Cell Lung Cancer	Phase 3	China
	NCT05384626 Tina Kehrig,clinicaltrials@nuvalent.com, 857-357-7000	A Study of NVL-655 in Patients With Advanced NSCLC and Other Solid Tumors Harboring ALK Rearrangement or Activating ALK Mutation (ALKOVE-1)	Phase 1 /Phase 2	Denver, CO; Philadelphia, PA; Saint Louis, MO; Stanford, CA; Chicago, IL; Columbus, OH; Sacramento, CA; Orange, CA; Baltimore, MD; Durham, NC; Seattle, WA; Detroit, MI; Miami, FL; Atlanta, GA; Washington, DC; Nashville, TN; New York, NY (2); Boston, MA (2); Belgium; Singapore (2); Japan (5); United Kingdom (4); Switzerland (2); Spain (5); Canada (4); Netherlands (2); Taiwan (3); Korea, Republic of (4); Italy (7); France (4); Australia (3); Germany (3)
	NCT06007937 Gregory Riely, MD, PhD,rielyg@mskcc.org,646-608-3913	A Study of Lorlatinib in Combination With Ramucirumab in People With Lung Cancer	Phase 1 /Phase 2	Harrison, NY; Montvale, NJ; Rockville Centre, NY; Commack, NY; New York, NY; Middletown, NJ
EML4-ALK Fusion	NCT03093116 BMS Study Connect Contact Center www.BMSStudyConnect.com,Clinical.Trials@bms.com,855-907-3286	A Study of Repotrectinib (TPX-0005) in Patients With Advanced Solid Tumors Harboring ALK, ROS1, or NTRK1-3 Rearrangements	Phase 1 /Phase 2	Appleton, WI; Saint Louis, MO; Baltimore, MD; Memphis, TN; Peoria, IL; Bolivar, MO; Fairfax, VA; Ann Arbor, MI; Goldsboro, NC; Philadelphia, PA; Chicago, IL; Columbus, OH; New Brunswick, NJ; Columbus, GA; Saint Paul, MN; Seattle, WA; Kingwood, TX; Cleveland, OH; Athens, GA; Hollywood, FL; Cincinnati, OH; Canton, OH; Dallas, TX; Tampa, FL; Houston, TX (2); Detroit, MI (2); Boston, MA (3); Washington, DC (2); Aurora, CO (2); New York, NY (3); CA (7); Korea, Democratic People's Republic of; Denmark; Singapore (2); Hungary (2); Hong Kong (2); Japan (9); United Kingdom (5); Spain (8); Canada (5); Netherlands (2); Belgium (2); China (30); Taiwan (3); Poland (5); Korea, Republic of (10); Italy (6); France (8); Australia (3); Germany (4)
	NCT05257512 Yinghui Sun, PhD,yhsun@centaurusbio.com,86-10-88858616	Phase I/II Study of SY-3505 in Patients With ALK-positive Advanced Non-small Cell Lung Cancer	Phase 1 /Phase 2	China
	NCT05341583 Chang Li Wang, Ph.D, wangchangli@tjmuch.com,86-22-23340123 x3180	Ensartinib as Adjuvant Treatment in Anaplastic Lymphoma Kinase (ALK) Positive Non-small Cell Lung Cancer	Phase 3	China
	NCT05384626 Tina Kehrig,clinicaltrials@nuvalent.com, 857-357-7000	A Study of NVL-655 in Patients With Advanced NSCLC and Other Solid Tumors Harboring ALK Rearrangement or Activating ALK Mutation (ALKOVE-1)	Phase 1 /Phase 2	Denver, CO; Philadelphia, PA; Saint Louis, MO; Stanford, CA; Chicago, IL; Columbus, OH; Sacramento, CA; Orange, CA; Baltimore, MD; Durham, NC; Seattle, WA; Detroit, MI; Miami,

This content is provided as a professional service and has not been reviewed or approved by the FDA

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
				FL; Atlanta, GA; Washington, DC; Nashville, TN; New York, NY (2); Boston, MA (2); Belgium; Singapore (2); Japan (5); United Kingdom (4); Switzerland (2); Spain (5); Canada (4); Netherlands (2); Taiwan (3); Korea, Republic of (4); Italy (7); France (4); Australia (3); Germany (3)
	NCT06007937 Gregory Riely, MD, PhD,rielyg@mskcc.org,646-608-3913	A Study of Lorlatinib in Combination With Ramucirumab in People With Lung Cancer	Phase 1 /Phase 2	Harrison, NY; Montvale, NJ; Rockville Centre, NY; Commack, NY; New York, NY; Middletown, NJ

This content is provided as a professional service and has not been reviewed or approved by the FDA

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
CDKN2A R80*	Abemaciclib	Verzenio	Cdk4/6/9 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (hormone receptor +, HER2-))
	Alrizomadlin		Small molecule inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor) Phase 2 (Melanoma, Salivary gland carcinoma, Acute myeloid leukemia (AML), Chronic myelomonocytic leukemia (CMML), Myelodysplastic Syndrome (MDS), Liposarcoma)
	ALRN-6924		Dual Mdm2/Mdm4 inhibitor, reactivates wild-type p53.	Phase 1 (Solid Tumor) Phase 1 (Cancer)
	Atirmociclib		Cdk4 inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Breast carcinoma)
	BGB-43395		Cdk4 inhibitor.	Phase 1 (Solid Tumor)
	Brigimadlin		Small molecule inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor)
	Dalpiciclib		Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Breast carcinoma)
	HDM201		Inhibitor of the Mdm2-p53 interaction.	Phase 1 (Solid Tumor) Phase 2 (Liposarcoma)
	Lerociclib		Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Breast carcinoma)
	Narazaciclib		Ark5 and Cdk4 multi-kinase inhibitor.	Phase 1 (Solid Tumor)
	Palbociclib	Ibrance	Cdk4/6 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (hormone receptor +, HER2-))
	PF-06873600		Cdk2/4/6 inhibitor.	Phase 2 (Ovarian carcinoma, Breast carcinoma)
	RGT-419B		CDK2/4/6 inhibitor.	Phase 1 (Solid Tumor)
	Ribociclib	Kisqali	Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (hormone receptor +, HER2-))
	SPH4336		Cdk4/6 inhibitor.	Phase 2 (Solid Tumor) Phase 3 (Breast carcinoma)
	STEMVAC		CD105 (ENG)/Yb-1/SOX2/CDH3 /MDM2 vaccine.	Phase 1 (Breast carcinoma)
	TQB3616		Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Breast carcinoma)
	Trilaciclib	Cosela	Cdk4/6 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Small cell lung carcinoma (SCLC))
	Voruciclib		Multikinase inhibitor of Cdk1/4/6/9.	Phase 1 (Solid Tumor) Phase 1 (Marginal zone lymphoma (MZL), Brain and Central Nervous System Tumors, Hematologic malignancies, Chronic lymphocytic leukemia (CLL), Acute myeloid leukemia (AML), Mantle cell lymphoma (MCL), Follicular lymphoma (FL), Diffuse large B-cell lymphoma (DLBCL), Small lymphocytic lymphoma)
EML4-ALK	Alkotinib		Dual Alk/Ros1 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))

This content is provided as a professional service and has not been reviewed or approved by the FDA

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
Fusion	Ensartinib		Alk small molecule kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	Entrectinib	Rozlytrek	Pan-Trk, Ros1, and Alk kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Solid tumor with NTRK fusion, ROS1-rearranged NSCLC)
	Iruplinalkib		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (ALK-rearranged NSCLC)
	Lorlatinib	Lorbrena	Alk/Ros1 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	NVL-520		Alk/Ros1 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Solid Tumor)
	NVL-655		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Solid Tumor)
	Repotrectinib	Augtyro	Alk/Ros1/Src/Trk/FAK kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Solid tumor with NTRK fusion, ROS1-rearranged NSCLC)
	SY-3505		Alk inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	TGRX-326		Alk/Ros1 inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	Unecritinib		Alk/Met/Ros1 kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Solid Tumor, Anaplastic large cell lymphoma (ALCL))
ALK L1196Q C1156Y	Alkotinib		Dual Alk/Ros1 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Ensartinib		Alk small molecule kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	Entrectinib	Rozlytrek	Pan-Trk, Ros1, and Alk kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Solid tumor with NTRK fusion, ROS1-rearranged NSCLC)
	Iruplinalkib		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (ALK-rearranged NSCLC)
	Lorlatinib	Lorbrena	Alk/Ros1 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (ALK-rearranged NSCLC)
	NVL-520		Alk/Ros1 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Solid Tumor)
	NVL-655		Alk inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Solid Tumor)
	Repotrectinib	Augtyro	Alk/Ros1/Src/Trk/FAK kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Solid tumor with NTRK fusion, ROS1-rearranged NSCLC)
	SY-3505		Alk inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	TGRX-326		Alk/Ros1 inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
ALK L1196Q	ALK was originally identified in anaplastic lymphoma as a fusion partner with the gene product of NPM1; ALK has subsequently been identified as a fusion partner with numerous other genes, including EML4 in lung cancer. ⁽¹⁾ The ALK gene can become oncogenic by a gene rearrangement, copy number gain, or genetic mutation. ^(1,2) Patients with EML4-ALK fusions generally have wild-type EGFR, KRAS, and TP53, and are resistant to Egfr inhibitors, although there have been reports of non-small cell lung carcinoma (NSCLC) tumors which harbor concomitant EGFR mutations and EML4-ALK translocations. ^(3,4) ALK rearrangements have been associated with younger age, increased disease stage, nodal metastasis, invasive mucinous disease, and with epithelial-mesenchymal transition marker expression in lung adenocarcinoma. ⁽⁵⁻¹⁰⁾	Tumors with ALK activation, by either mutation, fusion, or amplification, may be sensitive to Alk inhibitors. ⁽¹¹⁾ The Alk inhibitors crizotinib, ceritinib, alectinib, brigatinib, and lorlatinib have been approved for ALK-translocation-positive non-small cell lung cancer; additional Alk inhibitors are under investigation in clinical trials. ⁽¹²⁻¹⁷⁾ Crizotinib has additionally been FDA-approved for the treatment of pediatric and young adult patients with relapsed /refractory ALK-positive anaplastic large cell lymphoma or ALK-positive inflammatory myofibroblastic tumors. ⁽¹⁸⁻²⁰⁾ However, case studies have reported emergence of ALK L1196Q as a secondary mutation in ALK-rearranged patients following resistance to alectinib and crizotinib. ⁽²¹⁻²³⁾	Secondary mutations in ALK can confer resistance to different Alk inhibitors, such as crizotinib, ceritinib, alectinib, or brigatinib, in ALK-rearranged NSCLC. ⁽²⁴⁻²⁸⁾ Other mechanisms of resistance to crizotinib have also been reported, including activation of Egfr, K-Ras, PI3K/Akt /mTOR, and Hsp90 signaling. ⁽²⁹⁻³⁶⁾ Multiple case studies have reported that failure of alectinib treatment in NSCLC harboring ALK rearrangements is associated with transformation of disease into SCLC. ⁽³⁷⁻³⁹⁾
ALK C1156Y	ALK was originally identified in anaplastic lymphoma as a fusion partner with the gene product of NPM1; ALK has subsequently been identified as a fusion partner with numerous other genes, including EML4 in lung cancer. ⁽¹⁾ The ALK gene can become oncogenic by a gene rearrangement, copy number gain, or genetic mutation. ^(1,2) Patients with EML4-ALK fusions generally have wild-type EGFR, KRAS, and TP53, and are resistant to Egfr inhibitors, although there have been reports of non-small cell lung carcinoma (NSCLC) tumors which harbor concomitant EGFR mutations and EML4-ALK translocations. ^(3,4) ALK rearrangements have been associated with younger age, increased disease stage, nodal metastasis, invasive mucinous disease, and with epithelial-mesenchymal transition marker expression in lung adenocarcinoma. ⁽⁵⁻¹⁰⁾	Tumors with ALK activation, by either mutation, fusion, or amplification, may be sensitive to Alk inhibitors. The Alk inhibitors crizotinib, ceritinib, brigatinib, and lorlatinib have been approved for ALK-translocation-positive non-small cell lung cancer; additional Alk inhibitors are under investigation in clinical trials. ⁽¹²⁻¹⁷⁾ However, the mutation reported here has been reported to be resistant to ceritinib and crizotinib.	Secondary mutations in ALK can confer resistance to different Alk inhibitors, such as crizotinib, ceritinib, alectinib, or brigatinib, in ALK-rearranged NSCLC. ⁽²⁴⁻²⁸⁾ Other mechanisms of resistance to crizotinib have also been reported, including activation of Egfr, K-Ras, PI3K/Akt /mTOR, and Hsp90 signaling. ⁽²⁹⁻³⁶⁾ Multiple case studies have reported that failure of alectinib treatment in NSCLC harboring ALK rearrangements is associated with transformation of disease into SCLC. ⁽³⁷⁻³⁹⁾
ALK G1202L	ALK was originally identified in anaplastic lymphoma as a fusion partner with the gene product of NPM1; ALK has subsequently been identified as a fusion partner with numerous other genes, including EML4	Tumors with ALK activation, by either mutation, fusion, or amplification, may be sensitive to Alk inhibitors. ⁽¹¹⁾ The Alk inhibitors crizotinib, ceritinib, alectinib, brigatinib, and lorlatinib have been approved for ALK-translocation-	Secondary mutations in ALK can confer resistance to different Alk inhibitors, such as crizotinib, ceritinib, alectinib, or brigatinib, in ALK-rearranged NSCLC. ⁽²⁴⁻²⁸⁾ Other mechanisms of resistance to crizotinib

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
	in lung cancer. ⁽¹⁾ . The ALK gene can become oncogenic by a gene rearrangement, copy number gain, or genetic mutation. ^(1,2) . Patients with EML4-ALK fusions generally have wild-type EGFR, KRAS, and TP53, and are resistant to Egfr inhibitors, although there have been reports of non-small cell lung carcinoma (NSCLC) tumors which harbor concomitant EGFR mutations and EML4-ALK translocations. ^(3,4) . ALK rearrangements have been associated with younger age, increased disease stage, nodal metastasis, invasive mucinous disease, and with epithelial-mesenchymal transition marker expression in lung adenocarcinoma. ⁽⁵⁻¹⁰⁾ .	positive non-small cell lung cancer; additional Alk inhibitors are under investigation in clinical trials. ⁽¹²⁻¹⁷⁾ . However, a case study has reported the presence of G1202L in an NSCLC patient who harbored the EML4-ALK fusion and experienced disease progression after treatment with crizotinib followed by brigatinib; the patient subsequently experienced disease control for nine months following lorlatinib treatment. ⁽⁴⁰⁾ . In addition, clinical and preclinical findings have reported that another alteration at this position, ALK G1202R, is associated with resistance to crizotinib, ceritinib, alectinib, brigatinib, and entrectinib, but also sensitivity to lorlatinib, with additional preclinical findings suggesting that G1202R is associated with sensitivity to other Alk/Ros1 inhibitors like KRCA-0080 or JH-VIII-157-02. ^(27,30,41-48) .	have also been reported, including activation of Egfr, K-Ras, PI3K/Akt/mTOR, and Hsp90 signaling. ⁽²⁹⁻³⁶⁾ . Multiple case studies have reported that failure of alectinib treatment in NSCLC harboring ALK rearrangements is associated with transformation of disease into SCLC. ⁽³⁷⁻³⁹⁾ .
EML4-ALK Fusion	ALK was originally identified in anaplastic lymphoma as a fusion partner with the gene product of NPM1; ALK has subsequently been identified as a fusion partner with numerous other genes, including EML4 in lung cancer. ⁽¹⁾ . The ALK gene can become oncogenic by a gene rearrangement, copy number gain, or genetic mutation. ^(1,2) . Patients with EML4-ALK fusions generally have wild-type EGFR, KRAS, and TP53, and are resistant to Egfr inhibitors, although there have been reports of non-small cell lung carcinoma (NSCLC) tumors which harbor concomitant EGFR mutations and EML4-ALK translocations. ^(3,4) . ALK rearrangements have been associated with younger age, increased disease stage, nodal metastasis, invasive mucinous disease, and with epithelial-mesenchymal transition marker expression in lung adenocarcinoma. ⁽⁵⁻¹⁰⁾ .	Tumors with ALK activation, by either mutation, fusion, or amplification, may be sensitive to Alk inhibitors. ⁽¹¹⁾ . The Alk inhibitors crizotinib, ceritinib, alectinib, brigatinib, lorlatinib, and ensartinib have been approved for ALK-translocation-positive non-small cell lung cancer; additional Alk inhibitors are under investigation in clinical trials. ⁽¹²⁻¹⁷⁾ . Crizotinib has additionally been FDA-approved for the treatment of pediatric and young adult patients with relapsed/refractory ALK-positive anaplastic large cell lymphoma or ALK-positive inflammatory myofibroblastic tumors. ⁽¹⁸⁻²⁰⁾ . In this case, the presence of a crizotinib- and alectinib-resistance mutation suggests that this tumor may be resistant to these drugs. In this case, the presence of a crizotinib- and ceritinib-resistance mutation suggests that this tumor may be resistant to these drugs. In this case, the presence of an alectinib-, brigatinib-, ceritinib-, and crizotinib-resistance mutation suggests that this tumor may be resistant to these drugs.	Secondary mutations in ALK can confer resistance to different Alk inhibitors, such as crizotinib, ceritinib, alectinib, or brigatinib, in ALK-rearranged NSCLC. ⁽²⁴⁻²⁸⁾ . Other mechanisms of resistance to crizotinib have also been reported, including activation of Egfr, K-Ras, PI3K/Akt/mTOR, and Hsp90 signaling. ⁽²⁹⁻³⁶⁾ . Multiple case studies have reported that failure of alectinib treatment in NSCLC harboring ALK rearrangements is associated with transformation of disease into SCLC. ⁽³⁷⁻³⁹⁾ .
CDKN2A R80*	Because the CDKN2A and CDKN2B gene products encode proteins that act as tumor suppressors, deletion or loss of activity may result in deregulation of the p16INK4a/Cdk4/Cyclin/Rb and/or the Mdm2/p53 pathways, and altered regulation of the cell cycle. ^(49,50) . Loss of p16INK4a expression is thought to be an early	There are currently no drugs that directly target inactivating mutations or loss of CDKN2A. Because p16INK4a is known to inhibit Cdk4, tumors with CDKN2A alterations may be sensitive to Cdk4/6 inhibitors. ^(50,62) . p14ARF has been reported to function as a tumor suppressor through stabilization and activation of p53, via a mechanism	Inactivating CDKN2A alterations, most commonly deletion, have been significantly associated with resistance or lack of response to immune checkpoint blockade monotherapy in NSCLC patients; however, this association was not observed in

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
	event in NSCLC. ^(51,52) . One study reported that loss of CDKN2A and CDKN2B in the context of KRAS mutation accelerated the formation and progression of lung adenocarcinoma tumors in mice, as compared with KRAS mutation alone, resulting in high-grade tumors with lymph node metastasis. ⁽⁵³⁾ . Inactivating CDKN2A alterations or low mRNA expression has been significantly associated with brain metastasis in NSCLC patients. ⁽⁵⁴⁻⁵⁹⁾ . In addition, CDKN2A inactivation or low expression has been significantly associated with low expression of immune checkpoint inhibitor-related genes, including CD274/PD-L1, in NSCLC cases. ^(60,61) .	of Mdm2 inhibition. ^(50,64) . Trials of Mdm2 antagonists in solid tumors and hematological malignancies have shown clinical efficacy, but also some on-target toxicity. ⁽⁶⁵⁾ .	patients treated with immune checkpoint blockade plus chemotherapy. ^(57,61,66) .

References

1. Bagci O, Tumer S, Olgun N, Altungoz O "Copy number status and mutation analyses of anaplastic lymphoma kinase (ALK) gene in 90 sporadic neuroblastoma tumors." *Cancer letters*(2012): 72-7
2. Grande E, Bolós M, Arriola E "Targeting oncogenic ALK: a promising strategy for cancer treatment." *Molecular cancer therapeutics*(2011): 569-79
3. Tiseo M, Gelsomino F, Boggiani D, Bortesi B, Bartolotti M, Bozzetti C, Sammarelli G, Thai E, Ardizzoni A "EGFR and EML4-ALK gene mutations in NSCLC: a case report of erlotinib-resistant patient with both concomitant mutations." *Lung cancer (Amsterdam, Netherlands)*(2011): 241-3
4. Yang J, Zhang X, Su J, et al. "Concomitant EGFR mutation and EML4-ALK gene fusion in non-small cell lung cancer." *J Clin Oncol*(2011): Abstract 10517
5. Yang L, Ling Y, Guo L, Ma D, Xue X, Wang B, Li J, Ying J "Detection of ALK translocation in non-small cell lung carcinoma (NSCLC) and its clinicopathological significance using the Ventana immunohistochemical staining method: a single-center large-scale investigation of 1, 504 Chinese Han patients." *Chinese journal of cancer research = Chung-kuo yen cheng yen chiu*(2016): 495-502
6. Zhao M, Zhan C, Li M, Yang X, Yang X, Zhang Y, Lin M, Xia Y, Feng M, Wang Q "Aberrant status and clinicopathologic characteristic associations of 11 target genes in 1,321 Chinese patients with lung adenocarcinoma." *Journal of thoracic disease*(2018): 398-407
7. Possidente L, Landriscina M, Patitucci G, Borgia L, Lalinga V, Vita G "ALK rearrangement in specific subtypes of lung adenocarcinoma: immunophenotypic and morphological features." *Medical oncology (Northwood, London, England)*(2017): 76
8. Hu M, Zhang T, Yang Y, Che N, Li J, Liu Z, Li B "Driver gene alterations in lung adenocarcinoma: Demographic features of 2544 Chinese cases." *The International journal of biological markers*(2020): 44-50
9. Li D, Huang Y, Cai L, et al. "Genomic landscape of metastatic lung adenocarcinomas from large-scale clinical sequencing" *Neoplasia*(2021)
10. Kim H, Jang S, Chung D, Yoo S, Sun P, Jin Y, Nam K, Paik J, Chung J "A comprehensive comparative analysis of the histomorphological features of ALK-rearranged lung adenocarcinoma based on driver oncogene mutations: frequent expression of epithelial-mesenchymal transition markers than other genotype." *PloS one*(2013): e76999
11. Chand D, Yamazaki Y, Ruuth K, Schönherr C, Martinsson T, Kogner P, Attiyeh E, Maris J, Morozova O, Marra M, Ohira M, Nakagawara A, Sandström P, Palmer R, Hallberg B "Cell culture and Drosophila model systems define three classes of anaplastic lymphoma kinase mutations in neuroblastoma." *Disease models & mechanisms* (2013): 373-82
12. Solomon B, Shaw A, Ou S, et al. "OA 05.06 Phase 2 Study of Lorlatinib in Patients with Advanced ALK+/ROS1+ Non-Small-Cell Lung Cancer" *J Thorac Oncol*(2017)
13. Kwak E, Bang Y, Camidge D, Shaw A, Solomon B, Maki R, Ou S, Dezube B, Jänne P, Costa D, Varella-Garcia M, Kim W, Lynch T, Fidias P, Stubbs H, Engelman J, Sequist L, Tan W, Gandhi L, Mino-Kenudson M, Wei G, Shreeve S, Ratain M, Settleman J, Christensen J, Haber D, Wilner K, Salgia R, Shapiro G, Clark J, Iafrate A "Anaplastic lymphoma kinase inhibition in non-small-cell lung cancer." *The New England journal of medicine*(2010): 1693-703
14. Socinski M, Goldman J, El-Hariri I, Koczywas M, Vukovic V, Horn L, Paschold E, Salgia R, West H, Sequist L, Bonomi P, Brahmer J, Chen L, Sandler A, Belani C, Webb T, Harper H, Huberman M, Ramalingam S, Wong K, Teofilovici F, Guo W, Shapiro G "A multicenter phase II study of ganetespib monotherapy in patients with genotypically defined advanced non-small cell lung cancer." *Clinical cancer research : an official journal of the American Association for Cancer Research*(2013): 3068-77
15. Chabner B "Approval after phase I: ceritinib runs the three-minute mile." *The oncologist*(2014): 577-8
16. Ou S, Ahn J, De Petris L, Govindan R, Yang J, Hughes B, Lena H, Moro-Sibilot D, Bearz A, Ramirez S, Mekhail T, Spira A, Bordogna W, Balas B, Morcos P, Monnet A, Zeiter A, Kim D "Alectinib in Crizotinib-Refractory ALK-Rearranged Non-Small-Cell Lung Cancer: A Phase II Global Study." *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*(2016): 661-8
17. Gandhi L, Shaw A, Gadgeel SM et al. "A phase II, open-label, multicenter study of the ALK inhibitor alectinib in an ALK+ non-small-cell lung cancer (NSCLC) U.S. /Canadian population who had progressed on crizotinib (NP28761)." *J Clin Oncol*(2015)
18. Balis F, Thompson P, Mosse Y, Blaney S, Minard C, Weigel B, Fox E "First-dose and steady-state pharmacokinetics of orally administered crizotinib in children with solid tumors: a report on ADVL0912 from the Children's Oncology Group Phase 1/Pilot Consortium." *Cancer chemotherapy and pharmacology*(2017): 181-187
19. Mossé Y, Voss S, Lim M, Rolland D, Minard C, Fox E, Adamson P, Wilner K, Blaney S, Weigel B "Targeting ALK With Crizotinib in Pediatric Anaplastic Large Cell Lymphoma and Inflammatory Myofibroblastic Tumor: A Children's Oncology Group Study." *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*(2017): 3215-3221
20. Gambacorti-Passerini C, Orlov S, Zhang L, Braiteh F, Huang H, Esaki T, Horibe K, Ahn J, Beck J, Edenfield W, Shi Y, Taylor M, Tamura K, Van Tine B, Wu S, Paolini J, Selaru P, Kim T "Long-term effects of crizotinib in ALK-positive tumors (excluding NSCLC): A phase 1b open-label study." *American journal of hematology*(2018): 607-614
21. Zhang C, Wang Z, Zhuang R, Guo X, Feng Y, Shen F, Liu W, Zhang Y, Tong H, Sun W, Liu J, Wang G, Dai C, Lu W, Zhou Y "Efficacy and Resistance of ALK Inhibitors in Two Inflammatory Myofibroblastic Tumor Patients with ALK Fusions Assessed by Whole Exome and RNA Sequencing." *OncoTargets and therapy*(2020): 10335-10342
22. Singh H, Li Y, Spurr L, Shinagare A, Abhyankar R, Reilly E, Brais L, Nag A, Ducar M, Thorner A, Shapiro G, Keller R, Siletti C, Clark J, Farago A, Lin J, Demetri G, Gujrathi R, Kulke M, MacConaill L, Ligon A, Sicinska E, Meyerson M, Meyerhardt J, Cherniack A, Wolpin B, Ng K, Giannakis M, Hornick J, Cleary J "Molecular Characterization and Therapeutic Targeting of Colorectal Cancers Harboring Receptor Tyrosine Kinase Fusions." *Clinical cancer research : an official journal of the American Association for Cancer Research*(2021): 1695-1705
23. Furuta H, Araki M, Masago K, Sagae Y, Fujita S, Seto K, Shimizu J, Horio Y, Sasaki E, Hosoda W, Katayama R, Okuno Y, Hida T "Novel Resistance Mechanisms Including L1196Q, P1094H, and R1248_D1249 Insertion in Three Patients With NSCLC After ALK Tyrosine Kinase Inhibitor Treatment." *Journal of thoracic oncology : official publication of the International Association for the Study of Lung Cancer*(2021): 477-482
24. Toyokawa G, Hirai F, Inamasu E, Yoshida T, Nosaki K, Takenaka T, Yamaguchi M, Seto T, Takenoyama M, Ichinose Y "Secondary mutations at L1171 in the ALK gene confer resistance to both Crizotinib and Alectinib." *Journal of thoracic oncology : official publication of the International Association for the Study of Lung Cancer*(2014): e86-7
25. Cecccon M, Mologni L, Bisson W, Scapozza L, Gambacorti-Passerini C "Crizotinib-resistant NPM-ALK mutants confer differential sensitivity to unrelated Alk inhibitors." *Molecular cancer research : MCR*(2013): 122-32
26. Shaw A, Friboulet L, Leshchiner I, Gainer J, Bergqvist S, Brooun A, Burke B, Deng Y, Liu W, Dardaei L, Frias R, Schultz K, Logan J, James L, Smeal T, Timofeevski S, Katayama R, Iafrate A, Le L, McTigue M, Getz G, Johnson T, Engelman J "Resensitization to Crizotinib by the Lorlatinib ALK Resistance Mutation L1198F." *The New England journal of medicine*(2016): 54-61

References

27. Gainor J, Dardaei L, Yoda S, Friboulet L, Leshchiner I, Katayama R, Dagogo-Jack I, Gadgil S, Schultz K, Singh M, Chin E, Parks M, Lee D, DiCecca R, Lockerman E, Huynh T, Logan J, Ritterhouse L, Le L, Muniappan A, Digumarthy S, Channick C, Keyes C, Getz G, Dias-Santagata D, Heist R, Lennerz J, Sequist L, Benes C, Iafrate A, Mino-Kenudson M, Engelman J, Shaw A "Molecular Mechanisms of Resistance to First- and Second-Generation ALK Inhibitors in ALK-Rearranged Lung Cancer." *Cancer discovery*(2016): 1118-1133
28. Katayama R, Khan T, Benes C, Lifshits E, Ebi H, Rivera V, Shakespeare W, Iafrate A, Engelman J, Shaw A "Therapeutic strategies to overcome crizotinib resistance in non-small cell lung cancers harboring the fusion oncogene EML4-ALK." *Proceedings of the National Academy of Sciences of the United States of America*(2011): 7535-40
29. Sasaki T, Koivunen J, Ogino A, Yanagita M, Nikiforow S, Zheng W, Lathan C, Marcoux J, Du J, Okuda K, Capelletti M, Shimamura T, Ercan D, Stumpfova M, Xiao Y, Weremowicz S, Butaney M, Heon S, Wilner K, Christensen J, Eck M, Wong K, Lindeman N, Gray N, Rodig S, Jänne P "A novel ALK secondary mutation and EGFR signaling cause resistance to ALK kinase inhibitors." *Cancer research*(2011): 6051-60
30. Katayama R, Shaw A, Khan T, Mino-Kenudson M, Solomon B, Halmos B, Jessop N, Wain J, Yeo A, Benes C, Drew L, Saeh J, Crosby K, Sequist L, Iafrate A, Engelman J "Mechanisms of acquired crizotinib resistance in ALK-rearranged lung Cancers." *Science translational medicine*(2012): 120ra17
31. Yamaguchi N, Lucena-Araujo A, Nakayama S, de Figueiredo-Pontes L, Gonzalez D, Yasuda H, Kobayashi S, Costa D "Dual ALK and EGFR inhibition targets a mechanism of acquired resistance to the tyrosine kinase inhibitor crizotinib in ALK rearranged lung cancer." *Lung cancer (Amsterdam, Netherlands)*(2014): 37-43
32. Ji C, Zhang L, Cheng Y, Patel R, Wu H, Zhang Y, Wang M, Ji S, Belani C, Yang J, Ren X "Induction of autophagy contributes to crizotinib resistance in ALK-positive lung cancer." *Cancer biology & therapy*(2014): 570-7
33. Esfahani K, Agulnik J, Cohen V "A Systemic Review of Resistance Mechanisms and Ongoing Clinical Trials in ALK-Rearranged Non-Small Cell Lung Cancer." *Frontiers in oncology*(2014): 174
34. Cargnelli M, Corso S, Pergolizzi M, Mévellec L, Aisner D, Dziadziuszko R, Varella-Garcia M, Comoglio P, Doebele R, Vialard J, Giordano S "Activation of RAS family members confers resistance to ROS1 targeting drugs." *Oncotarget*(2015): 5182-94
35. Rossing H, Grauslund M, Urbanska E, Melchior L, Rask C, Costa J, Skov B, Sørensen J, Santoni-Rugiu E "Concomitant occurrence of EGFR (epidermal growth factor receptor) and KRAS (V-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog) mutations in an ALK (anaplastic lymphoma kinase)-positive lung adenocarcinoma patient with acquired resistance to crizotinib: a case report." *BMC research notes*(2013): 489
36. Doebele R, Pilling A, Aisner D, Kutateladze T, Le A, Weickhardt A, Kondo K, Linderman D, Heasley L, Franklin W, Varella-Garcia M, Camidge D "Mechanisms of resistance to crizotinib in patients with ALK gene rearranged non-small cell lung cancer." *Clinical cancer research : an official journal of the American Association for Cancer Research*(2012): 1472-82
37. Miyamoto S, Ikushima S, Ono R, Awano N, Kondo K, Furuhashi Y, Fukumoto K, Kumasaka T "Transformation to small-cell lung cancer as a mechanism of acquired resistance to crizotinib and alectinib." *Japanese journal of clinical oncology*(2016): 170-3
38. Fujita S, Masago K, Katakami N, Yatabe Y "Transformation to SCLC after Treatment with the ALK Inhibitor Alectinib." *Journal of thoracic oncology : official publication of the International Association for the Study of Lung Cancer*(2016): e67-72
39. Takegawa N, Hayashi H, Iizuka N, Takahama T, Ueda H, Tanaka K, Takeda M, Nakagawa K "Transformation of ALK rearrangement-positive adenocarcinoma to small-cell lung cancer in association with acquired resistance to alectinib." *Annals of oncology : official journal of the European Society for Medical Oncology*(2016): 953-5
40. Meng Z, Li T, Wang P, Lizaso A, Huang D "The efficacy of lorlatinib in a lung adenocarcinoma patient with a novel ALK G1202L mutation: a case report." *Cancer biology & therapy*(2021): 1-4
41. Ignatius Ou S, Azada M, Hsiang D, Herman J, Kain T, Siwak-Tapp C, Casey C, He J, Ali S, Klempner S, Miller V "Next-generation sequencing reveals a Novel NSCLC ALK F1174V mutation and confirms ALK G1202R mutation confers high-level resistance to alectinib (CH5424802/RO5424802) in ALK-rearranged NSCLC patients who progressed on crizotinib." *Journal of thoracic oncology : official publication of the International Association for the Study of Lung Cancer*(2014): 549-53
42. Ardini E, Menichincheri M, Banfi P, Bosotti R, De Ponti C, Pulci R, Ballinari D, Ciomei M, Texido G, Degrossi A, Avanzi N, Amboldi N, Saccardo M, Casero D, Orsini P, Bandiera T, Mologni L, Anderson D, Wei G, Harris J, Vernier J, Li G, Felder E, Donati D, Isacchi A, Pesenti E, Magnaghi P, Galvani A "Entrectinib, a Pan-TRK, ROS1, and ALK Inhibitor with Activity in Multiple Molecularly Defined Cancer Indications." *Molecular cancer therapeutics*(2016): 628-39
43. Fontana D, Ceccon M, Gambacorti-Passerini C, Mologni L "Activity of second-generation ALK inhibitors against crizotinib-resistant mutants in an NPM-ALK model compared to EML4-ALK." *Cancer medicine*(2015): 953-65
44. Hatcher J, Bahcall M, Choi H, Gao Y, Sim T, George R, Jänne P, Gray N "Discovery of Inhibitors That Overcome the G1202R Anaplastic Lymphoma Kinase Resistance Mutation." *Journal of medicinal chemistry*(2015): 9296-9308
45. Kang C, Yun J, Lee K, Lee H, Yun C, Hwang J, Cho S, Jung H, Kim P, Ha J, Jeon J, Choi S, Jeong H, Kim H, Park C "Development of potent ALK inhibitor and its molecular inhibitory mechanism against NSCLC harboring EML4-ALK proteins." *Biochemical and biophysical research communications*(2015): 762-7
46. Zhang S, Anjum R, Squillace R, Nadworny S, Zhou T, Keats J, Ning Y, Wardwell S, Miller D, Song Y, Eichinger L, Moran L, Huang W, Liu S, Zou D, Wang Y, Mohammad Q, Jang H, Ye E, Narasimhan N, Wang F, Miret J, Zhu X, Clackson T, Dalgarno D, Shakespeare W, Rivera V "The Potent ALK Inhibitor Brigatinib (AP26113) Overcomes Mechanisms of Resistance to First- and Second-Generation ALK Inhibitors in Preclinical Models." *Clinical cancer research : an official journal of the American Association for Cancer Research*(2016): 5527-5538
47. Dagogo-Jack I, Brannon A, Ferris L, Campbell C, Lin J, Schultz K, Ackil J, Stevens S, Dardaei L, Yoda S, Hubbeling H, Digumarthy S, Riester M, Hata A, Sequist L, Lennes I, Iafrate A, Heist R, Azzoli C, Farago A, Engelman J, Lennerz J, Benes C, Leary R, Shaw A, Gainor J "Tracking the Evolution of Resistance to ALK Tyrosine Kinase Inhibitors through Longitudinal Analysis of Circulating Tumor DNA." *JCO precision oncology*(2018)
48. Solomon BJ, Bauer TM, Felip E, et al. "Safety and efficacy of lorlatinib (PF-06463922) from the dose-escalation component of a study in patients with advanced ALK+ or ROS1+ non-small cell lung cancer (NSCLC)." *J Clin Oncol*(2016): Abstract 9009
49. Gazzeri S, Gouyer V, Vour'ch C, Brambilla C, Brambilla E "Mechanisms of p16INK4A inactivation in non small-cell lung cancers." *Oncogene*(1998): 497-504
50. Zhang Y, Xiong Y, Yarbrough W "ARF promotes MDM2 degradation and stabilizes p53: ARF-INK4a locus deletion impairs both the Rb and p53 tumor suppression pathways." *Cell*(1998): 725-34
51. Kratzke R, Greatens T, Rubins J, Maddaus M, Niewoehner D, Niehans G, Geradts J "Rb and p16INK4a expression in resected non-small cell lung tumors." *Cancer research*(1996): 3415-20
52. Kawabuchi B, Moriyama S, Hironaka M, Fujii T, Koike M, Moriyama H, Nishimura Y, Mizuno S, Fukayama M "p16 inactivation in small-sized lung adenocarcinoma: its association with poor prognosis." *International journal of cancer*(1999): 49-53

References

53. Schuster K, Venkateswaran N, Rabellino A, Girard L, Peña-Llopis S, Scaglioni P "Nullifying the CDKN2AB locus promotes mutant K-ras lung tumorigenesis." Molecular cancer research : MCR(2014): 912-23
54. Deng Z, Cui L, Li P, Ren N, Zhong Z, Tang Z, Wang L, Gong J, Cheng H, Guan Y, Yi X, Xia X, Zhou R, He Z "Genomic comparison between cerebrospinal fluid and primary tumor revealed the genetic events associated with brain metastasis in lung adenocarcinoma." Cell death & disease(2021): 935
55. Huang R, Harries L, Decker B, Hiemenz M, Murugesan K, Creeden J, Tolba K, Stabile L, Ramkissoon S, Burns T, Ross J "Clinicopathologic and Genomic Landscape of Non-Small Cell Lung Cancer Brain Metastases." The oncologist(2022): 839-848
56. Lin Y, Wang Y, Yeh D, Hung C, Yeh Y, Ho H, Mon H, Chen M, Wu Y, Chou T "Gene Expression Profile in Primary Tumor Is Associated with Brain-Tropism of Metastasis from Lung Adenocarcinoma." International journal of molecular sciences(2021)
57. Gutiontov S, Turchan W, Spurr L, Rouhani S, Chervin C, Steinhardt G, Lager A, Wanjari P, Malik R, Connell P, Chmura S, Juloori A, Hoffman P, Ferguson M, Donington J, Patel J, Vokes E, Weichselbaum R, Bestvina C, Segal J, Pitroda S "CDKN2A loss-of-function predicts immunotherapy resistance in non-small cell lung cancer." Scientific reports(2021): 20059
58. Lengel H, Mastrogiacomio B, Connolly J, Tan K, Liu Y, Fick C, Dunne E, He D, Lankadasari M, Satravada B, Sun Y, Kundra R, Fong C, Smith S, Riely G, Rudin C, Gomez D, Solit D, Berger M, Li B, Mayo M, Matei I, Lyden D, Adusumilli P, Schultz N, Sanchez-Vega F, Jones D "Genomic mapping of metastatic organotropism in lung adenocarcinoma." Cancer cell(2023): 970-985.e3
59. Lara-Mejía L, Cardona A, Mas L, Martin C, Samtani S, Corrales L, Cruz-Rico G, Remon J, Galvez-Nino M, Ruiz R, Rios-Garcia E, Tejada F, Lozano-Vazquez N, Rosell R, Arrieta O "Impact of Concurrent Genomic Alterations on Clinical Outcomes in Patients With ALK-Rearranged NSCLC." Journal of thoracic oncology : official publication of the International Association for the Study of Lung Cancer(2024): 119-129
60. Zhu L, Wang Z, Sun Y, Giamas G, Stebbing J, Yu Z, Peng L "A Prediction Model Using Alternative Splicing Events and the Immune Microenvironment Signature in Lung Adenocarcinoma." Frontiers in oncology(2021): 778637
61. Ebot E, Duncan D, Tolba K, Fabrizio D, Frampton G, Comment L, Albacker L "Deletions on 9p21 are associated with worse outcomes after anti-PD-1/PD-L1 monotherapy but not chemoimmunotherapy." NPJ precision oncology(2022): 44
62. Stone S, Jiang P, Dayananth P, Tavtigian S, Katcher H, Parry D, Peters G, Kamb A "Complex structure and regulation of the P16 (MTS1) locus." Cancer research(1995): 2988-94
63. Sherr C, Bertwistle D, DEN Besten W, Kuo M, Sugimoto M, Tago K, Williams R, Zindy F, Roussel M "p53-Dependent and -independent functions of the Arf tumor suppressor." Cold Spring Harbor symposia on quantitative biology(2005): 129-37
64. Ozenne P, Eymin B, Brambilla E, Gazzeri S "The ARF tumor suppressor: structure, functions and status in cancer." International journal of cancer(2010): 2239-47
65. Khoo K, Verma C, Lane D "Drugging the p53 pathway: understanding the route to clinical efficacy." Nature reviews. Drug discovery(2014): 217-36
66. Zhao L, Zhou X, Li H, Yin T, Jiang Y "Prognosis of immunotherapy for non-small cell lung cancer with CDKN2A loss of function." Journal of thoracic disease(2024): 507-515

REPORTING	PHYSICIAN
Report Date: JAN-18-2025	Susanne Arnold
Receipt Date: JAN-11-2025	University of KY HNR
Collection Date: JAN-09-2025	800 Rose St, Ben Roach Bldg CC 222, Lexington, KY, 40508, United States
Specimen: Blood	Ph: (859) 257-4488 Fax: (859) 257-0332
Status: FINAL	Additional Recipient: N/A

Companion Diagnostic

No reportable alterations with companion diagnostic (CDx) claims
--

Other Biomarkers Identified

Results reported in this section are not prescriptive or conclusive for labeled use of any specific therapeutic product. See professional services section for additional information.

Biomarkers with Evidence of Clinical Significance in Tissue and ctDNA [†]

Biomarker	Status	Additional Information
ALK fusions	DETECTED	EML4-ALK Fusion See professional services section for additional information

[†]Please refer below to Performance Characteristics and Definitions sections for descriptions of categories.

Other Biomarkers with Potential Clinical Significance

Clinical significance has not yet been established for biomarkers in this section. See the professional services section for additional information.

- ALK C1156Y
- ALK L1196Q
- CDKN2A R80*

Intended Use

Guardant360® CDx is a qualitative next generation sequencing-based in vitro diagnostic device that uses targeted high throughput hybridization-based capture technology for detection of single nucleotide variants (SNVs), insertions and deletions (indels) in 55 genes, copy number amplifications (CNAs) in two (2) genes, and fusions in four (4) genes. Guardant360 CDx utilizes circulating cell-free DNA (cfDNA) from plasma of peripheral whole blood collected in Streck Cell-Free DNA Blood Collection Tubes (BCTs). The test is intended to be used as a companion diagnostic to identify patients who may benefit from treatment with the therapies listed in **Table 1** in accordance with the approved therapeutic product labeling.

Table 1. Companion Diagnostic Indications

Indication	Biomarker	Therapy
Non-small cell lung cancer (NSCLC)	EGFR exon 19 deletions, L858R, and T790M*	TAGRISSO® (osimertinib)
	EGFR exon 20 insertions	RYBREVENT® (amivantamab-vmjw)
	ERBB2/HER2 activating mutations (SNVs and exon 20 insertions)	ENHERTU® (fam-trastuzumab deruxtecan-nxki)
	KRAS G12C	LUMAKRAS™ (sotorasib)
Breast cancer	ESR1 missense mutations between codons 310 and 547	ORSERDU™ (elacestrant)

A negative result from a plasma specimen does not assure that the patient's tumor is negative for genomic findings. Patients who are negative for the biomarkers listed in **Table 1** should be reflexed to tissue biopsy testing for **Table 1** biomarkers using an FDA-approved tumor tissue test, if feasible.

*The efficacy of TAGRISSO (osimertinib) has not been established in the *EGFR* T790M plasma-positive, tissue-negative or unknown population and clinical data for T790M plasma-positive patients are limited; therefore, testing using plasma specimens is most appropriate for consideration in patients from whom a tumor biopsy cannot be obtained.

Additionally, the test is intended to provide tumor mutation profiling to be used by qualified health care professionals in accordance with professional guidelines in oncology for cancer patients with any solid malignant neoplasm. The test is for use with patients previously diagnosed with cancer and in conjunction with other laboratory and clinical findings.

Genomic findings other than those listed in **Table 1** are not prescriptive or conclusive for labeled use of any specific therapeutic product.

Guardant360 CDx is a single-site assay performed at Guardant Health, Inc.

Warnings and Precautions

- Alterations reported may include somatic (not inherited) or germline (inherited) alterations. The assay filters germline variants from reporting except for pathogenic *BRCA1*, *BRCA2*, *ATM*, and *CDK12* alterations. However, if a reported alteration is suspected to be germline, confirmatory testing should be considered in the appropriate clinical context.
- The test is not intended to replace germline testing or to provide information about cancer predisposition.
- Somatic alterations in *ATM* and *CDK12* are not reported by the test as they are excluded from the test's reportable range.
- Genomic findings from cfDNA may originate from circulating tumor DNA (ctDNA) fragments, germline alterations, or non-tumor somatic alterations, such as clonal hematopoiesis of indeterminate potential (CHIP).
- Allow the tube to fill completely until blood stops flowing into the tube. Underfilling of tubes with less than 5 mL of blood (bottom of the label indicates 5 mL fill when tube is held vertically) may lead to incorrect analytical results or poor product performance. This tube has been designed to fill with 10 mL of blood.

Limitations

- For *in vitro* diagnostic use.
- For prescription use only. This test must be ordered by a qualified medical professional in accordance with clinical laboratory regulations.
- The efficacy of TAGRISSO (osimertinib) has not been established in the *EGFR* T790M plasma-positive, tissue-negative or unknown population, and clinical data for T790M plasma-positive patients are limited; therefore, testing using plasma specimens is most appropriate for consideration in patients from whom a tumor biopsy cannot be obtained.
- TAGRISSO efficacy has not been established in patients with *EGFR* exon 19 deletions < 0.08% MAF, in patients with *EGFR* L858R < 0.09% MAF, and in patients with *EGFR* T790M < 0.03% MAF.
- RYBREVENT efficacy has not been established in patients with *EGFR* exon 20 insertions < 0.02% MAF.
- LUMAKRAS efficacy has not been established in patients with *KRAS* G12C biomarkers < 0.11% MAF.
- ENHERTU efficacy has not been established in patients with *ERBB2* exon 20 insertions < 0.03% MAF and in patients with *ERBB2* SNVs < 0.23% MAF.
- ORSERDU efficacy has not been established in patients with *ESR1* missense mutations < 0.03% MAF.
- The test is not intended to be used for standalone diagnostic purposes.
- The test is intended to be performed on specific serial number-controlled instruments by Guardant Health, Inc.
- A negative result for any given variant does not preclude the presence of this variant in tumor tissue.
- Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care.
- ctDNA shedding rate may be lower in patients with primary central nervous system (CNS) tumors.

Performance Characteristics

Please refer to product label, www.guardant360cdx.com/technicalinfo. Clinical Performance has not been established for biomarkers in categories 2, 3A, 3B and 4. Guardant360 CDx is indicated to report the following SNVs (*AKT1*, *ALK*, *APC*, *AR*, *ARAF*, *ATM*#, *BRAF*, *BRCA1*##, *BRCA2*##, *CCND1*, *CDH1*, *CDK4*, *CDK6*, *CDK12*#, *CDKN2A*, *CTNNB1*, *EGFR*, *ERBB2*, *ESR1*, *FGFR1*, *FGFR2*, *FGFR3*, *GATA3*, *GNA11*, *GNAQ*, *HRAS*, *IDH1*, *IDH2*, *KIT*, *KRAS*, *MAP2K1*, *MAP2K2*, *MET*, *MLH1*, *MTOR*, *MYC*, *NF1*, *NFE2L2*, *NRAS*, *NTRK1*, *NTRK3*, *PDGFRA*, *PIK3CA*, *PTEN*, *RAF1*, *RET*, *RHEB*, *ROS1*, *SMAD4*, *SMO*, *STK11*, *TERT*, *TSC1*, *VHL*), Indels (*ALK*, *AKT1*, *APC*, *ATM*#, *BRAF*, *BRCA1*##, *BRCA2*##, *CDH1*, *CDK12*#, *CDKN2A*, *EGFR*, *ERBB2*, *ESR1*, *FGFR2*, *GATA3*, *HNFI1A*, *HRAS*, *KIT*, *KRAS*, *MET*, *MLH1*, *NF1*, *PDGFRA*, *PIK3CA*, *PTEN*, *RET*, *ROS1*, *STK11*, *TSC1*, *VHL*), Fusion (*ALK*, *NTRK1*, *RET*, *ROS1*), and Amplifications (*ERBB2*, *HER2*).

#Reporting is enabled for pathogenic germline alterations only. Somatic alterations will not be reported. | ##Reporting is enabled for both germline and somatic alterations.

Definition of Categories
The test report includes genomic finding reported in the following categories:

Category	Prescriptive use for Therapeutic Product	Clinical Performance	Analytical Performance	Comments
Category 1: Companion Diagnostic (CDx)	Yes	Yes	Yes	ctDNA biomarkers linked to the safe and effective use of the corresponding therapeutic product, for which Guardant360 CDx has demonstrated clinical performance shown to support therapeutic efficacy and strong analytical performance for the biomarker.
Category 2: ctDNA Biomarkers with Strong Evidence of Clinical Significance in ctDNA	No	No	Yes	ctDNA biomarkers with strong evidence of clinical significance presented by other FDA-approved liquid biopsy companion diagnostics for which Guardant360 CDx has demonstrated analytical reliability but not clinical performance.
Category 3A: Biomarkers with Evidence of Clinical Significance in tissue supported by: strong analytical validation using ctDNA	No	No	Yes	ctDNA biomarkers with evidence of clinical significance presented by tissue-based FDA-approved companion diagnostics or professional guidelines for which Guardant360 CDx has demonstrated analytical performance including analytical accuracy, and concordance of blood-based testing to tissue-based testing for the biomarker.
Category 3B: Biomarkers with Evidence of Clinical Significance in tissue supported by: analytical validation using ctDNA	No	No	Yes	ctDNA biomarkers with evidence of clinical significance presented by tissue-based FDA-approved companion diagnostics or professional guidelines for which Guardant360 CDx has demonstrated minimum analytical performance including analytical accuracy.
Category 4: Other Biomarkers with Potential Clinical Significance	No	No	Yes	ctDNA biomarkers with emergent evidence based on peer-reviewed publications for genes/variants in tissue, variant information from well- curated public databases, or <i>in-vitro</i> pre-clinical models, for which Guardant360 CDx has demonstrated minimum analytical performance.

Testing performed at: Guardant Health
Laboratory Director: Martina Lefterova, MD PhD | CLIA ID: 05D2070300 | CAP #: 8765297 | 505 Penobscot Drive Redwood City, CA, 94063, USA